Simple rag results :

Model	BertScore (%)	Faithfulness (%)	Answer relevancy(%)	Context precision(%)	Context recall(%)	Overall (%)
qwen/qwen3-vl-8b-instruct	86.57	81.62	79.83	83.13	79.83	**82.19**
google/gemma-3-12b-it	**86.96**	**81.63**	74.47	**83.19**	81.15	81.46
mistralai/ministral-14b-2512	82.89	75.25	**80.27**	83.13	**83.38**	80.98

TABLE 4.13 – Naive RAG RAGAS and similarity Results

Qwen3-vl-8b-instruct offers the most balanced profile with the highest overall performance (82.19). Gemma-3-12b-it achieved the highest BERTScore (86.96), indicating answers semantically closest to the ground truth, together with the highest Faithfulness (81.63), indicating that the statements in the generated response are almost entirely supported by the retrieved context with minimal hallucination. Ministral-14b obtains the best Answer relevancy (80.27) and Context recall (83.38), showing that it addresses the user's question most directly while covering most of the reference facts ; however, this comes at the cost of the lowest BERTScore and faithfulness, suggesting a less extractive answering style. Context precision is statistically indistinguishable across models (83.1383.19) : since all configurations share the same retrieval stack, the retrieved chunks are identical by construction, and as this RAGAS metric judges chunk usefulness with respect to the generated answer, its residual variation only reflects minor differences in how each model exploited that common context.

2. Agentic RAG results :
Two phases : agentic reasoning (routing) and generation(synthesis)

- **Routing Evaluation Results**

Routing Model	Synthesis Model	Routing Accuracy(%)	Routing Precision(%)	Routing Recall(%)	Routing F1(%)
google/gemma-4-31b-it	google/gemma-3-12b-it	**85.71**	**88.86**	84.44	**84.67**
google/gemma-4-31b-it	qwen/qwen3-vl-8b-instruct	**85.71**	86.41	**84.56**	84.61
google/gemma-4-31b-it	mistralai/ministral-14b-2512	**85.71**	88.40	84.55	84.95
google/gemma-4-26b-a4b-it	mistralai/ministral-14b-2512	82.80	86.16	82.42	82.00
google/gemma-4-26b-a4b-it	google/gemma-3-12b-it	81.31	83.68	80.33	79.61
google/gemma-4-26b-a4b-it	qwen/qwen3-vl-8b-instruct	80.21	83.94	79.86	79.86

- Similarity and RAGAS results table

Routing Model	Synthesis Model	BertScore (%)	Faithfulness (%)	Answer relevancy (%)	Context precision (%)	Context recall (%)	Overall (%)
google/gemma-4-26b-a4b-it	qwen/qwen3-vl-8b-instruct	85.18	93.58	79.90	**93.53**	81.38	**86.71**
google/gemma-4-26b-a4b-it	mistralai/ministral-14b-2512	84.37	87.02	**80.48**	93.38	**86.95**	86.44
google/gemma-4-31b-it	qwen/qwen3-vl-8b-instruct	**87.03**	89.89	76.19	91.81	85.43	86.07
google/gemma-4-26b-a4b-it	google/gemma-3-12b-it	85.44	**95.51**	68.27	90.67	85.45	85.07
google/gemma-4-31b-it	google/gemma-3-12b-it	86.48	93.55	65.79	89.37	84.69	83.98
google/gemma-4-31b-it	mistralai/ministral-14b-2512	85.00	83.06	71.67	91.54	84.04	83.06

TABLE 4.20 – Agentic RAG Similarity and RAGAS Evaluation (ranked by overall perfomance)

Best overall (production-safe) configuration : gemma-4-31b-it (router) with gemma3-12b-it (synthesis). This pairing achieves a routing accuracy of 85.71% (tied for best) and is the only configuration combining perfect out-of-scope rejection (refusal recall = 100%) with the highest refusal precision (90.0%, implied refusal F1 ≈94.7%), alongside near-top faithfulness (93.55%, third-highest and within 0.03pt of second place). Its main weakness is answer relevancy, which falls to 65.79% the lowest value observed across all six agentic RAG configurations, a direct consequence of gemma-3-12b-it's extractive, source-hugging answering style and its Overall RAGAS/similarity score (83.98) is second-lowest of the six congurations. This is the safest default when refusal correctness matters more than answer engagement, though relevancy would benet from targeted prompt tuning if adopted.